In [1]:
```
# Primary VAR (VAR-SPY): Tables 3, 4 and 5, Figure 2, Tables B.1, B.3, C.1 and C.2, Figures
D.1 and D.2
import os
import warnings
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.vector_ar.var_model import VAR

warnings.filterwarnings("ignore")
os.makedirs("output", exist_ok=True)

# Plot style for the thesis figures
plt.rcParams.update({
    "font.family": "serif", "font.serif": ["Times New Roman", "Liberation Serif", "DejaVu
Serif"],
    "mathtext.fontset": "dejavuserif", "font.size": 10,
    "axes.spines.top": False, "axes.spines.right": False, "axes.axisbelow": True,
    "axes.grid": True, "axes.grid.axis": "y", "grid.color": "#e6e5e0", "grid.linewidth": 0.6,
    "legend.frameon": False, "savefig.dpi": 300,
})
blue, orange, green, grey, red = "#2a78d6", "#eb6834", "#1baf7a", "#52514e", "#8b1a1a"
```

In [2]:
```
# Merge the clean files into one monthly data set, December 2015 to December 2025 (121
months)
files = ["spy_clean.csv", "rsp_clean.csv", "voo_clean.csv", "vix_clean.csv",
"dgs10_clean.csv",
         "AIStocks_ASVI8_Monthly.csv", "AIStocks_ASVI4_Monthly.csv",
         "mag7_clean.csv", "mag7b_clean.csv", "mag7c_clean.csv"]
df = pd.read_csv(files[0], parse_dates=["date"])
for f in files[1:]:
    df = df.merge(pd.read_csv(f, parse_dates=["date"]), on="date", how="inner")
df = df.sort_values("date").reset_index(drop=True)

# 2021-2025 subsample. December 2020 only serves as the first lag, so the VARs run on Jan
2021 to Dec 2025.
sub = df[df["date"] >= "2020-12-31"].reset_index(drop=True)
print(len(df), "months,", df["date"].min().date(), "to", df["date"].max().date())

labels = {"dlog_vix": "dlogVIX", "d_dgs10": "dDGS10", "ASVI8": "ASVI8", "ASVI4": "ASVI4",
          "flow_spy": "Flow SPY", "flow_rsp": "Flow RSP", "flow_voo": "Flow VOO",
          "mag7_ret": "Mag7 returns", "mag7b_ret": "Mag7B returns", "mag7c_ret": "Mag7C
returns"}

# Cholesky order: VIX, yields, attention, ETF flow, Mag7 returns
cols = ["dlog_vix", "d_dgs10", "ASVI8", "flow_spy", "mag7_ret"]
samples = {"Full sample": df, "2021-2025": sub}
```

```
121 months, 2015-12-31 to 2025-12-31
```

In [3]:
```
def stars(p):
    return "***" if p < 0.01 else "**" if p < 0.05 else "*" if p < 0.10 else ""

def granger(res, cause, effect):
    # F-test: do the lags of `cause` help predict `effect`?
    t = res.test_causality(effect, [cause], kind="f")
    return float(t.test_statistic), float(t.pvalue)


def fevd(res, response):
    # Share (%) of the forecast error variance of `response` explained by each shock, h = 1,
6, 12
    d = res.fevd(12).decomp[list(res.names).index(response)] * 100
    return pd.DataFrame({"h=1": d[0], "h=6": d[5], "h=12": d[11]}, index=list(res.names))


def irf_bands(res, impulse, response, h=12):
    # Orthogonalized impulse response with asymptotic 95% bands
    i, j = list(res.names).index(response), list(res.names).index(impulse)
    irf = res.irf(h)
    point = irf.orth_irfs[:, i, j]
    se = irf.stderr(orth=True)[:, i, j]
```

```
    return point, point - 1.959964 * se, point + 1.959964 * se


def largest_eigenvalue(res):
    # statsmodels reports the roots of the characteristic polynomial. The eigenvalues of the
    # companion matrix are their inverses, so the largest eigenvalue is 1 / smallest root.
    return 1 / np.abs(res.roots).min()
```

In [4]:

```
# Table B.1: lag selection, VAR(0) to VAR(4) on the same estimation sample. * marks the
minimum.
rows = []
for name, data in samples.items():
    so = VAR(data[cols]).select_order(maxlags=4)
    best = so.selected_orders
    print(name, {k: int(v) for k, v in best.items()})
    for p in range(5):
        rows.append({"Sample": name, "Lag order": p,
                     "AIC": f"{so.ics['aic'][p]:.2f}" + ("*" if best["aic"] == p else ""),
                     "BIC": f"{so.ics['bic'][p]:.2f}" + ("*" if best["bic"] == p else ""),
                     "FPE (x 10^-9)": f"{so.ics['fpe'][p] * 1e9:.3f}" + ("*" if best["fpe"]
== p else ""),
                     "HQIC": f"{so.ics['hqic'][p]:.2f}" + ("*" if best["hqic"] == p else
"")})
tableB1 = pd.DataFrame(rows)
tableB1.to_csv("output/Table_B1_lag_selection.csv", index=False)
tableB1
```

```
Full sample {'aic': 2, 'bic': 1, 'hqic': 1, 'fpe': 2}
2021-2025 {'aic': 1, 'bic': 1, 'hqic': 1, 'fpe': 1}
```

Out[4]:

	Sample	Lag order	AIC	BIC	FPE (x 10^-9)	HQIC
0	Full sample	0	-19.77	-19.66	2.585	-19.73
1	Full sample	1	-20.65	-19.95*	1.072	-20.37*
2	Full sample	2	-20.67*	-19.37	1.056*	-20.14
3	Full sample	3	-20.49	-18.60	1.270	-19.73
4	Full sample	4	-20.36	-17.88	1.461	-19.36
5	2021-2025	0	-19.55	-19.37	3.229	-19.48
6	2021-2025	1	-20.61*	-19.54*	1.120*	-20.20*
7	2021-2025	2	-20.59	-18.62	1.167	-19.83
8	2021-2025	3	-20.43	-17.56	1.449	-19.32
9	2021-2025	4	-20.56	-16.80	1.411	-19.10

In [5]:

```
# VAR(1), lag chosen by BIC in both samples
res = {name: VAR(data[cols]).fit(1) for name, data in samples.items()}
for name, r in res.items():
    print(name)
    print(r.summary())
```

Full sample
  Summary of Regression Results
==================================
Model: VAR
Method: OLS
Date: Sat, 10, Oct, 2026
Time: 15:00:15

No. of Equations: 5.00000   BIC: -20.0111
Nobs: 120.000   HQIC: -20.4250
Log likelihood: 421.118   FPE: 1.01579e-09
AIC: -20.7080   Det(Omega_mle):   7.95900e-10

Results for equation dlog_vix
==
                 coefficient   std. error   t-stat   prob
----
const   -0.011007   0.022832   -0.482   0.630
L1.dlog_vix   -0.179885   0.104036   -1.729   0.084
L1.d_dgs10   -0.039471   0.085615   -0.461   0.645
L1.ASVI8   -0.011769   0.038549   -0.305   0.760
L1.flow_spy   0.880865   0.797564   1.104   0.269
L1.mag7_ret   0.525984   0.334003   1.575   0.115
==

Results for equation d_dgs10
==
                 coefficient   std. error   t-stat   prob
----
const   0.016786   0.025088   0.669   0.503
L1.dlog_vix   -0.085124   0.114319   -0.745   0.457
L1.d_dgs10   0.099845   0.094076   1.061   0.289
L1.ASVI8   -0.023546   0.042359   -0.556   0.578
L1.flow_spy   0.472412   0.876390   0.539   0.590
L1.mag7_ret   0.120816   0.367013   0.329   0.742
==

Results for equation ASVI8
==
                 coefficient   std. error   t-stat   prob
----
const   0.038451   0.034718   1.108   0.268
L1.dlog_vix   -0.016115   0.158198   -0.102   0.919
L1.d_dgs10   -0.098276   0.130186   -0.755   0.450
L1.ASVI8   0.771495   0.058618   13.161   0.000
L1.flow_spy   0.296174   1.212779   0.244   0.807
L1.mag7_ret   0.663194   0.507886   1.306   0.192
==

Results for equation flow_spy
==
                 coefficient   std. error   t-stat   prob
----
const   0.002186   0.002700   0.810   0.418
L1.dlog_vix   -0.004352   0.012304   -0.354   0.724
L1.d_dgs10   0.009145   0.010126   0.903   0.366
L1.ASVI8   -0.005725   0.004559   -1.256   0.209
L1.flow_spy   -0.236425   0.094327   -2.506   0.012
L1.mag7_ret   -0.010470   0.039502   -0.265   0.791
==

Results for equation mag7_ret
==
                 coefficient   std. error   t-stat   prob
----
const   0.019817   0.007557   2.622   0.009
L1.dlog_vix   -0.035176   0.034436   -1.021   0.307
L1.d_dgs10   -0.014767   0.028339   -0.521   0.602
L1.ASVI8   0.027440   0.012760   2.151   0.032
L1.flow_spy   0.007849   0.263994   0.030   0.976
L1.mag7_ret   -0.125510   0.110555   -1.135   0.256
==

Correlation matrix of residuals

```
            dlog_vix   d_dgs10     ASVI8  flow_spy  mag7_ret
dlog_vix    1.000000  0.033670 -0.048826 -0.264349 -0.535061
d_dgs10     0.033670  1.000000 -0.044692  0.022269 -0.172120
ASVI8      -0.048826 -0.044692  1.000000  0.037966  0.169103
flow_spy   -0.264349  0.022269  0.037966  1.000000  0.207487
mag7_ret   -0.535061 -0.172120  0.169103  0.207487  1.000000
```

2021-2025

```
  Summary of Regression Results
==================================
Model:                         VAR
Method:                        OLS
Date:           Sat, 10, Oct, 2026
Time:                     15:00:15
--------------------------------------------------------------------
No. of Equations:         5.00000    BIC:                   -19.4744
Nobs:                     60.0000    HQIC:                  -20.1120
Log likelihood:           219.966    FPE:                1.22756e-09
AIC:                     -20.5216    Det(Omega_mle):     7.62219e-10
--------------------------------------------------------------------
Results for equation dlog_vix
================================================================================
                 coefficient       std. error           t-stat            prob
--------------------------------------------------------------------------------
const               0.009465         0.030801            0.307           0.759
L1.dlog_vix        -0.438256         0.137139           -3.196           0.001
L1.d_dgs10         -0.075866         0.109816           -0.691           0.490
L1.ASVI8           -0.055372         0.042572           -1.301           0.193
L1.flow_spy        -1.148259         1.217238           -0.943           0.346
L1.mag7_ret         0.299292         0.440679            0.679           0.497
================================================================================

Results for equation d_dgs10
================================================================================
                 coefficient       std. error           t-stat            prob
--------------------------------------------------------------------------------
const               0.052784         0.042516            1.242           0.214
L1.dlog_vix         0.069576         0.189298            0.368           0.713
L1.d_dgs10          0.122080         0.151583            0.805           0.421
L1.ASVI8           -0.059401         0.058763           -1.011           0.312
L1.flow_spy         1.268928         1.680194            0.755           0.450
L1.mag7_ret         0.646280         0.608284            1.062           0.288
================================================================================

Results for equation ASVI8
================================================================================
                 coefficient       std. error           t-stat            prob
--------------------------------------------------------------------------------
const               0.045578         0.057697            0.790           0.430
L1.dlog_vix        -0.326832         0.256889           -1.272           0.203
L1.d_dgs10         -0.022676         0.205708           -0.110           0.912
L1.ASVI8            0.803656         0.079745           10.078           0.000
L1.flow_spy        -0.622650         2.280125           -0.273           0.785
L1.mag7_ret         0.543043         0.825479            0.658           0.511
================================================================================

Results for equation flow_spy
================================================================================
                 coefficient       std. error           t-stat            prob
--------------------------------------------------------------------------------
const               0.003591         0.003437            1.045           0.296
L1.dlog_vix         0.006590         0.015303            0.431           0.667
L1.d_dgs10          0.001908         0.012254            0.156           0.876
L1.ASVI8           -0.004186         0.004750           -0.881           0.378
L1.flow_spy        -0.327822         0.135825           -2.414           0.016
L1.mag7_ret         0.029611         0.049173            0.602           0.547
================================================================================

Results for equation mag7_ret
================================================================================
                 coefficient       std. error           t-stat            prob
--------------------------------------------------------------------------------
```

```
const               0.008378         0.011658            0.719           0.472
L1.dlog_vix         0.004374         0.051905            0.084           0.933
L1.d_dgs10         -0.007533         0.041564           -0.181           0.856
L1.ASVI8            0.037788         0.016113            2.345           0.019
L1.flow_spy        -0.182225         0.460706           -0.396           0.692
L1.mag7_ret        -0.112385         0.166790           -0.674           0.500
================================================================================

Correlation matrix of residuals
            dlog_vix   d_dgs10     ASVI8  flow_spy  mag7_ret
dlog_vix    1.000000  0.341317 -0.069740 -0.308788 -0.511279
d_dgs10     0.341317  1.000000  0.002385 -0.244646 -0.415818
ASVI8      -0.069740  0.002385  1.000000 -0.155951  0.196066
flow_spy   -0.308788 -0.244646 -0.155951  1.000000  0.157979
mag7_ret   -0.511279 -0.415818  0.196066  0.157979  1.000000
```

In [6]:

```
# Stability and residual tests
for name, r in res.items():
    w = r.test_whiteness(nlags=10)
    n = r.test_normality()
    print(f"{name}: largest eigenvalue {largest_eigenvalue(r):.3f} (smallest root
{np.abs(r.roots).min():.3f}), "
          f"Portmanteau Q {w.test_statistic:.1f} (critical value {w.crit_value:.1f}, p
{w.pvalue:.3f}), "
          f"normality p {n.pvalue:.3f}")
```

```
Full sample: largest eigenvalue 0.793 (smallest root 1.261), Portmanteau Q 234.9 (critical
value 261.0, p 0.312), normality p 0.000
2021-2025: largest eigenvalue 0.838 (smallest root 1.194), Portmanteau Q 221.0 (critical value
261.0, p 0.562), normality p 0.000
```

In [7]:

```
# Table 3: Mag7 returns equation
regressors = ["const"] + ["L1." + c for c in cols]
names = {"const": "Constant", **{"L1." + c: "L1." + labels[c] for c in cols}}
table3 = {}
for name, r in res.items():
    col = []
    for v in regressors:
        col += [f"{r.params.loc[v, 'mag7_ret']:.4f}{stars(r.pvalues.loc[v, 'mag7_ret'])}",
                f"({r.stderr.loc[v, 'mag7_ret']:.4f})"]
    col += [int(r.nobs), f"{r.llf:.1f}", f"{r.aic:.2f}", f"{r.bic:.2f}", "Yes" if
r.is_stable() else "No",
            f"{r.test_whiteness(nlags=10).pvalue:.3f}"]
    table3[name] = col
index = [x for v in regressors for x in (names[v], "")] + [
    "Observations", "Log-likelihood (system)", "AIC (system)", "BIC (system)", "Stability",
"Portmanteau p-value"]
table3 = pd.DataFrame(table3, index=index)
table3.to_csv("output/Table_3_VAR_SPY_Mag7_equation.csv")
table3
```

Out[7]:

	Full sample	2021-2025
Constant	0.0198***	0.0084
	(0.0076)	(0.0117)
L1.dlogVIX	-0.0352	0.0044
	(0.0344)	(0.0519)
L1.dDGS10	-0.0148	-0.0075
	(0.0283)	(0.0416)
L1.ASVI8	0.0274**	0.0378**
	(0.0128)	(0.0161)
L1.Flow SPY	0.0078	-0.1822
	(0.2640)	(0.4607)
L1.Mag7 returns	-0.1255	-0.1124
	(0.1106)	(0.1668)
Observations	120	60
Log-likelihood (system)	421.1	220.0
AIC (system)	-20.71	-20.52
BIC (system)	-20.01	-19.47
Stability	Yes	Yes
Portmanteau p-value	0.312	0.562

In [8]:

```
# Table 4: Granger causality
tests = [("ASVI8", "mag7_ret", "ASVI8 does not cause Mag7 returns"),
         ("ASVI8", "flow_spy", "ASVI8 does not cause Flow SPY"),
         ("flow_spy", "mag7_ret", "Flow SPY does not cause Mag7 returns"),
         ("mag7_ret", "ASVI8", "Mag7 returns do not cause ASVI8"),
         ("dlog_vix", "mag7_ret", "dlogVIX does not cause Mag7 returns"),
         ("d_dgs10", "mag7_ret", "dDGS10 does not cause Mag7 returns")]
rows = []
for cause, effect, label in tests:
    row = {"Null hypothesis": label}
    for name, r in res.items():
        f, p = granger(r, cause, effect)
        row[f"F ({name})"] = f"{f:.3f}{stars(p)}"
        row[f"p ({name})"] = f"{p:.3f}"
    rows.append(row)
table4 = pd.DataFrame(rows)
table4.to_csv("output/Table_4_Granger_causality.csv", index=False)
table4
```

Out[8]:

	Null hypothesis	F (Full sample)	p (Full sample)	F (2021-2025)	p (2021-2025)
0	ASVI8 does not cause Mag7 returns	4.625**	0.032	5.500**	0.020
1	ASVI8 does not cause Flow SPY	1.577	0.210	0.776	0.379
2	Flow SPY does not cause Mag7 returns	0.001	0.976	0.156	0.693
3	Mag7 returns do not cause ASVI8	1.705	0.192	0.433	0.511
4	dlogVIX does not cause Mag7 returns	1.043	0.307	0.007	0.933
5	dDGS10 does not cause Mag7 returns	0.272	0.603	0.033	0.856

In [9]:

```
# Table 5: variance decomposition of Mag7 returns (%)
fevd_mag7 = {name: fevd(r, "mag7_ret") for name, r in res.items()}
table5 = pd.concat({name: f.rename(index=labels) for name, f in fevd_mag7.items()},
axis=1).round(1).astype(object)

# Last row: ASVI8 share divided by the Flow SPY share at h = 12
table5.loc["ASVI8 / Flow SPY"] = ""
for name, f in fevd_mag7.items():
    ratio = f.loc["ASVI8", "h=12"] / f.loc["flow_spy", "h=12"]
    table5.loc["ASVI8 / Flow SPY", (name, "h=12")] = f"{ratio:.0f}x" if ratio >= 20 else f"
{ratio:.1f}x"
```

```
table5.to_csv("output/Table_5_FEVD_Mag7.csv")
table5
```

Out[9]:

	Full sample			2021-2025		
	h=1	h=6	h=12	h=1	h=6	h=12
dlogVIX	28.6	27.7	27.7	26.1	24.7	24.4
dDGS10	2.4	2.4	2.4	6.6	6.0	5.9
ASVI8	1.9	4.6	4.8	2.8	9.8	10.9
Flow SPY	0.5	0.5	0.5	0.0	0.2	0.2
Mag7 returns	66.6	64.8	64.6	64.4	59.3	58.5
ASVI8 / Flow SPY			9.9x			47x

In [10]:

```
# Impulse responses (Section 4.6), one-standard-deviation shocks
for name, r in res.items():
    point, lower, upper = irf_bands(r, "ASVI8", "mag7_ret")
    shock = np.linalg.cholesky(r.sigma_u)[2, 2]                    # size of a one-SD ASVI8
shock
    print(f"{name}: one-SD ASVI8 shock = {shock:.2f}")
    print(f"  Mag7 response to ASVI8 (%): t=0 {point[0] * 100:.2f}, t=1 {point[1] * 100:.2f},
"
          f"sum t=1 to 12 {point[1:].sum() * 100:.1f}, band excludes zero at t =",
          np.where((lower > 0) | (upper < 0))[0].tolist())
    point, lower, upper = irf_bands(r, "dlog_vix", "mag7_ret")
    print(f"  Mag7 response to dlogVIX (%): t=0 {point[0] * 100:.2f}, band excludes zero at t
=",
          np.where((lower > 0) | (upper < 0))[0].tolist())

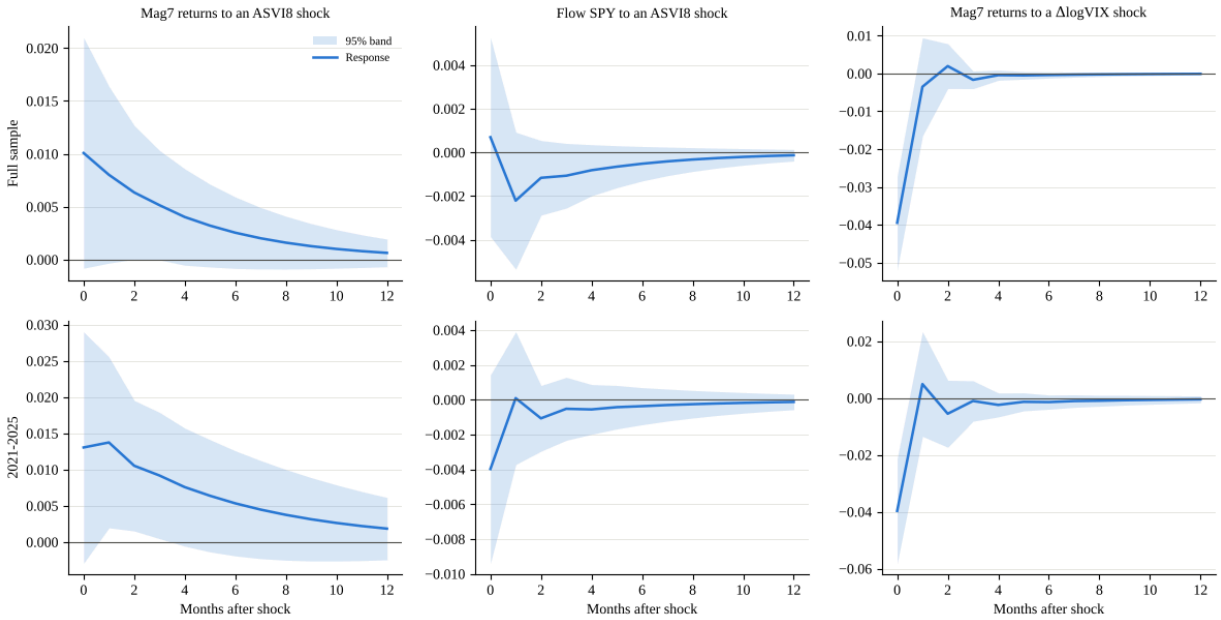
growth = res["2021-2025"].params.loc["L1.ASVI8", "mag7_ret"] / res["Full
sample"].params.loc["L1.ASVI8", "mag7_ret"] - 1
print(f"\nASVI8 coefficient in 2021-2025 vs full sample: {growth:+.0%}")
```

```
Full sample: one-SD ASVI8 shock = 0.34
  Mag7 response to ASVI8 (%): t=0 1.01, t=1 0.80, sum t=1 to 12 3.7, band excludes zero at t =
[2]
  Mag7 response to dlogVIX (%): t=0 -3.95, band excludes zero at t = [0]
2021-2025: one-SD ASVI8 shock = 0.38
  Mag7 response to ASVI8 (%): t=0 1.31, t=1 1.37, sum t=1 to 12 7.1, band excludes zero at t =
[1, 2, 3]
  Mag7 response to dlogVIX (%): t=0 -3.97, band excludes zero at t = [0]

ASVI8 coefficient in 2021-2025 vs full sample: +38%
```

In [11]:

```
# Figure 2: the three responses discussed in Sections 4.6.1 to 4.6.3 (top: full sample,
bottom: 2021-2025)
panels = [("ASVI8", "mag7_ret", "Mag7 returns to an ASVI8 shock"),
          ("ASVI8", "flow_spy", "Flow SPY to an ASVI8 shock"),
          ("dlog_vix", "mag7_ret", r"Mag7 returns to a $\Delta$logVIX shock")]
fig, axes = plt.subplots(2, 3, figsize=(12, 6.2))
for row, (name, r) in enumerate(res.items()):
    for col, (impulse, response, title) in enumerate(panels):
        ax = axes[row, col]
        point, lower, upper = irf_bands(r, impulse, response)
        ax.fill_between(range(13), lower, upper, color=blue, alpha=0.18, lw=0, label="95%
band")
        ax.plot(range(13), point, color=blue, lw=1.8, label="Response")
        ax.axhline(0, color=grey, lw=0.8)
        ax.set_xticks(range(0, 13, 2))
        if row == 0:
            ax.set_title(title, fontsize=10)
        if row == 1:
            ax.set_xlabel("Months after shock")
        if col == 0:
            ax.set_ylabel(name)
axes[0, 0].legend(fontsize=8, loc="upper right")
plt.tight_layout()
fig.savefig("output/Figure_2_key_IRFs.png", dpi=300, bbox_inches="tight")
plt.show()
```



In [12]:
```
# Table B.3: residual correlations (lower triangle)
for name, r in res.items():
    corr = pd.DataFrame(np.corrcoef(np.asarray(r.resid).T),
                        index=[labels[c] for c in cols], columns=[labels[c] for c in cols])
    lower = np.tril(np.ones(corr.shape, dtype=bool))
    corr = corr.apply(lambda c: c.map(lambda x: f"{x:.3f}")).where(lower, "")
    corr.to_csv(f"output/Table_B3_residual_correlations_{name.replace(' ', '_')}.csv")
    print(name)
    print(corr, "\n")
```

```
Full sample
             dlogVIX  dDGS10  ASVI8 Flow SPY Mag7 returns
dlogVIX        1.000
dDGS10         0.034   1.000
ASVI8         -0.049  -0.045  1.000
Flow SPY      -0.264   0.022  0.038    1.000
Mag7 returns  -0.535  -0.172  0.169    0.207        1.000

2021-2025
             dlogVIX  dDGS10   ASVI8 Flow SPY Mag7 returns
dlogVIX        1.000
dDGS10         0.341   1.000
ASVI8         -0.070   0.002   1.000
Flow SPY      -0.309  -0.245  -0.156    1.000
Mag7 returns  -0.511  -0.416   0.196    0.158        1.000
```

In [13]:
```
# Tables C.1 and C.2: all five equations, OLS standard errors in parentheses
def equation_r2(r):
    y = np.asarray(r.endog)[r.k_ar:]
    e = np.asarray(r.resid)
    return 1 - (e ** 2).sum(axis=0) / ((y - y.mean(axis=0)) ** 2).sum(axis=0)

for name, r in res.items():
    table = {}
    for i, eq in enumerate(cols):
        col = []
        for v in regressors:
            col += [f"{r.params.loc[v, eq]:.4f}{stars(r.pvalues.loc[v, eq])}", f"
({r.stderr.loc[v, eq]:.4f})"]
        col += [f"{equation_r2(r)[i]:.3f}", int(r.nobs)]
        table["Eq.: " + labels[eq]] = col
    table = pd.DataFrame(table, index=[x for v in regressors for x in (names[v], "")] +
["R2", "Observations"])
    number = "C1" if name == "Full sample" else "C2"
    table.to_csv(f"output/Table_{number}_VAR_SPY_all_equations_{name.replace(' ', '_')}.csv")
    print(f"{name}: log-likelihood {r.llf:.1f}, AIC {r.aic:.2f}, BIC {r.bic:.2f}")
    display(table)
```

Full sample: log-likelihood 421.1, AIC -20.71, BIC -20.01

	Eq.: dlogVIX	Eq.: dDGS10	Eq.: ASVI8	Eq.: Flow SPY	Eq.: Mag7 returns
Constant	-0.0110	0.0168	0.0385	0.0022	0.0198***
	(0.0228)	(0.0251)	(0.0347)	(0.0027)	(0.0076)
L1.dlogVIX	-0.1799*	-0.0851	-0.0161	-0.0044	-0.0352
	(0.1040)	(0.1143)	(0.1582)	(0.0123)	(0.0344)
L1.dDGS10	-0.0395	0.0998	-0.0983	0.0091	-0.0148
	(0.0856)	(0.0941)	(0.1302)	(0.0101)	(0.0283)
L1.ASVI8	-0.0118	-0.0235	0.7715***	-0.0057	0.0274**
	(0.0385)	(0.0424)	(0.0586)	(0.0046)	(0.0128)
L1.Flow SPY	0.8809	0.4724	0.2962	-0.2364**	0.0078
	(0.7976)	(0.8764)	(1.2128)	(0.0943)	(0.2640)
L1.Mag7 returns	0.5260	0.1208	0.6632	-0.0105	-0.1255
	(0.3340)	(0.3670)	(0.5079)	(0.0395)	(0.1106)
R2	0.120	0.028	0.642	0.074	0.046
Observations	120	120	120	120	120

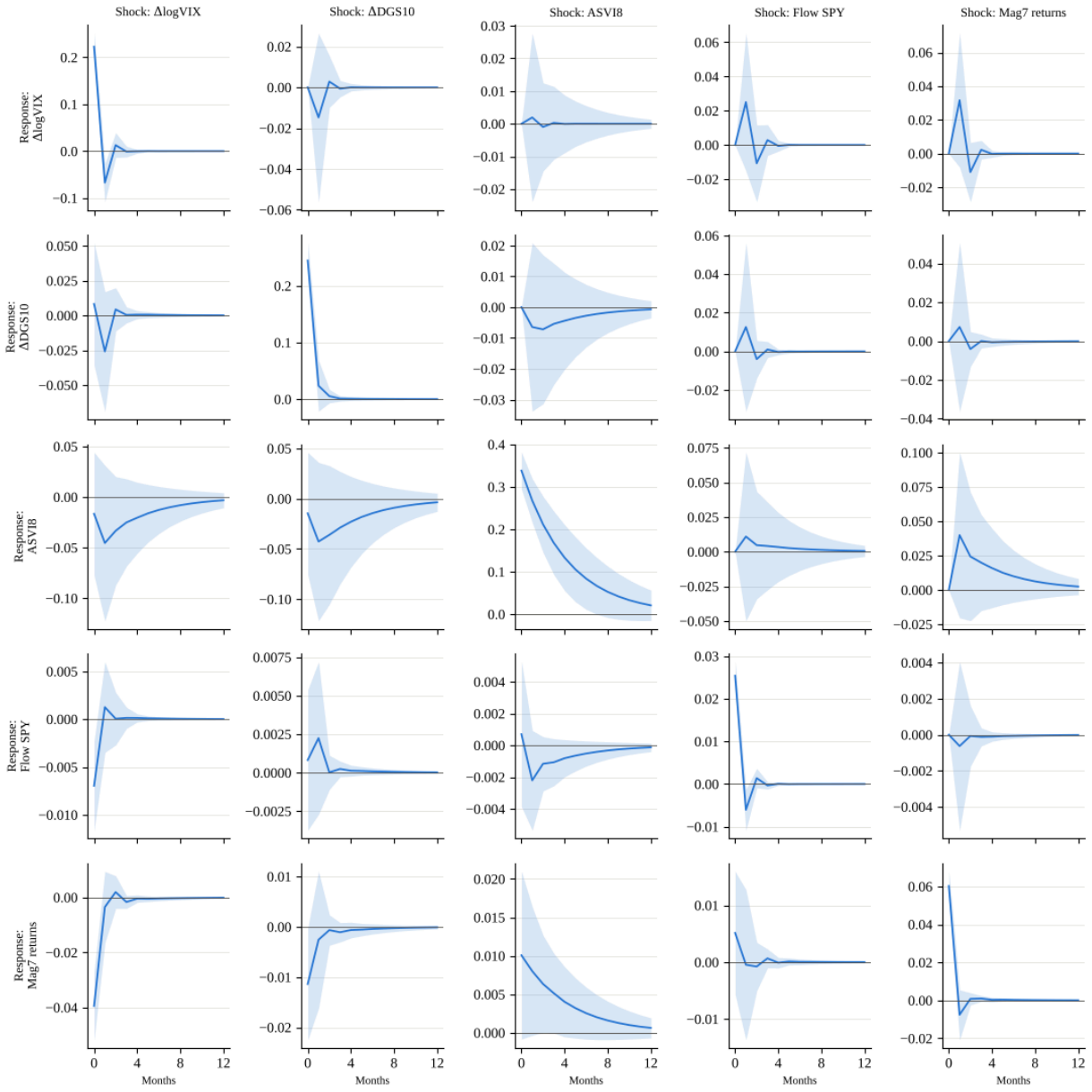
2021-2025: log-likelihood 220.0, AIC -20.52, BIC -19.47

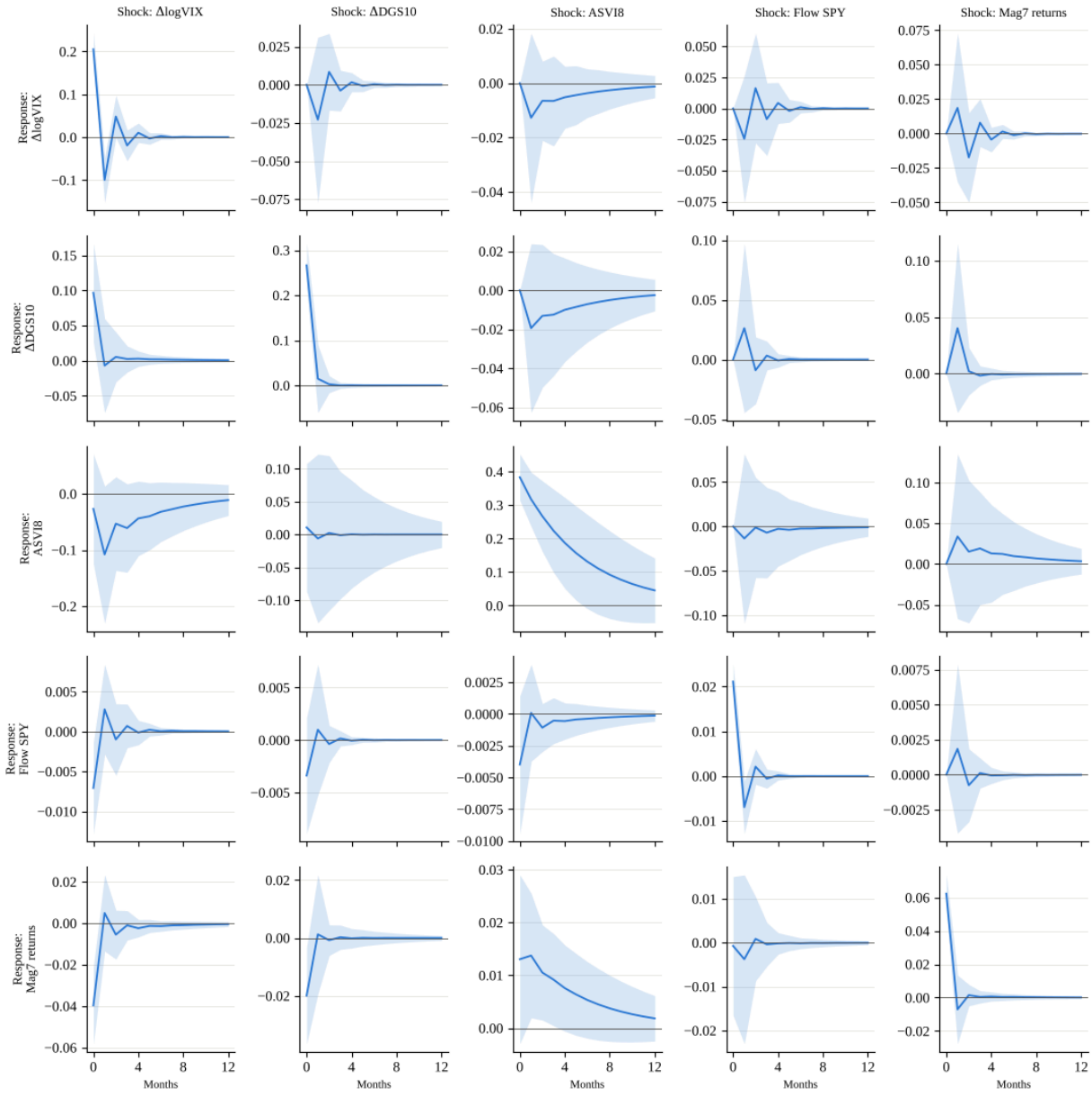
	Eq.: dlogVIX	Eq.: dDGS10	Eq.: ASVI8	Eq.: Flow SPY	Eq.: Mag7 returns
Constant	0.0095	0.0528	0.0456	0.0036	0.0084
	(0.0308)	(0.0425)	(0.0577)	(0.0034)	(0.0117)
L1.dlogVIX	-0.4383***	0.0696	-0.3268	0.0066	0.0044
	(0.1371)	(0.1893)	(0.2569)	(0.0153)	(0.0519)
L1.dDGS10	-0.0759	0.1221	-0.0227	0.0019	-0.0075
	(0.1098)	(0.1516)	(0.2057)	(0.0123)	(0.0416)
L1.ASVI8	-0.0554	-0.0594	0.8037***	-0.0042	0.0378**
	(0.0426)	(0.0588)	(0.0797)	(0.0048)	(0.0161)
L1.Flow SPY	-1.1483	1.2689	-0.6226	-0.3278**	-0.1822
	(1.2172)	(1.6802)	(2.2801)	(0.1358)	(0.4607)
L1.Mag7 returns	0.2993	0.6463	0.5430	0.0296	-0.1124
	(0.4407)	(0.6083)	(0.8255)	(0.0492)	(0.1668)
R2	0.268	0.045	0.711	0.119	0.105
Observations	60	60	60	60	60

In [14]:

```
# Figures D.1 and D.2: all 25 impulse responses (rows: responses, columns: shocks)
plot_labels = dict(labels, dlog_vix=r"$\Delta$logVIX", d_dgs10=r"$\Delta$DGS10")
for name, r in res.items():
    irf = r.irf(12)
    point, se = irf.orth_irfs, irf.stderr(orth=True)
    fig, axes = plt.subplots(5, 5, figsize=(11, 11), sharex=True)
    for i in range(5):
        for j in range(5):
            ax = axes[i, j]
            ax.fill_between(range(13), point[:, i, j] - 1.959964 * se[:, i, j],
                            point[:, i, j] + 1.959964 * se[:, i, j], color=blue, alpha=0.18,
lw=0)
            ax.plot(range(13), point[:, i, j], color=blue, lw=1.3)
            ax.axhline(0, color=grey, lw=0.6)
            if i == 0:
                ax.set_title("Shock: " + plot_labels[cols[j]], fontsize=9)
            if j == 0:
                ax.set_ylabel("Response:\n" + plot_labels[cols[i]], fontsize=9)
            if i == 4:
                ax.set_xticks([0, 4, 8, 12])
                ax.set_xlabel("Months", fontsize=8)
    plt.tight_layout()
    fig.savefig("output/Figure_D1_IRF_full_sample.png" if name == "Full sample" else
"output/Figure_D2_IRF_2021_2025.png",
```

```
                dpi=300, bbox_inches="tight")
plt.show()
```




Shock: ΔlogVIX
Shock: ΔDGS10
Shock: ASVI8
Shock: Flow SPY
Shock: Mag7 returns
Response: ΔlogVIX
Response: ΔDGS10
Response: ASVI8
Response: Flow SPY
Response: Mag7 returns
Months